# FOX commercial inverter
# Modbus interface definition description

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# 1 Supported model list

This section describes the inverter models that use this protocol, and the minimum firmware version requirements required for these models. When the host computer connects to the inverter, please pay attention to the FW version.

## 1.1 Model description

**Table 1-1 List of** supported model series and firmware version requirements

| Model series | Series ID | Minimum firmware version requirements |
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# 2 Introduction

The Modbus protocol is issued as a general device-level communication protocol standard. This document describes the Modbus protocol for FoxESS inverter, which is used to standardie subsequent third-party integrated development. Since the FoxESS inverter complies with the standard modbus specification, this document focuses on the information specific to the FoxESS inverter.

For additional information, please refer to the description in the ModBus standard specification document. For a detailed description of the standard protocols used by the FoxESS Inverter as well as customized sections and examples of interactions, please read **5 Communication Protocol Overview** .

## 2.1 Terms and abbreviations

**Table 2-1** Terms and abbreviation definitions

| Name | Description |
|---|---|
| master node | In master-slave communication, the party that actively initiates communication is called the master node. |
| slave node | In master-slave communication, the party that passively responds to commands is called a slave node. |
| broadcast address | Fixed to 0 |
| Register address | The register address corresponds to a 2-byte message |
| U8 | Unsigned 8-bit integer |
| U16 | Unsigned 16-bit integer |
| U32 | Unsigned 32-bit integer |
| I8 | Signed 8-bit integer |
| I16 | Signed 16-bit integer |
| I32 | Signed 32-bit integer |
| STR | string |
| MLD | multibyte |
| Bitfield16 | 16-bit wide bitwise representation of data |
| Bitfield32 | 32-bit wide bitwise representation of data |
| - | not involving |
| s | Second |
| INV | inverter |
| BMS | battery management system |
| | |

| Name | Description |
|---|---|
| RO | Read-only data |
| RW | Readable and writable data |
| WO | Write data only |

# 3 Register definition

**Table 3-1 Inverter model definition table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 1 | Model name | RO | STR | N/A | 1 | 30000 | 16 | |
| 2 | SN | RO | STR | N/A | 1 | 30016 | 16 | |
| 3 | MFG ID | RO | STR | N/A | 1 | 30032 | 16 | |

**Table 3-2 Inverter version definition table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 4 | Master Version | RO | U16 | N/A | N/A | 36001 | 1 | |
| 5 | Slave Version | RO | U16 | N/A | N/A | 36002 | 1 | |
| 6 | Manager Version | RO | U16 | N/A | N/A | 36003 | 1 | |
| 7 | Meter1 SN | RO | STR | N/A | N/A | 36100 | 16 | |
| 8 | Meter1 MFG ID | RO | STR | N/A | N/A | 36116 | 16 | |
| 9 | Meter1 TYPE | RO | STR | N/A | N/A | 36132 | 16 | |
| 10 | Meter1 Version | RO | STR | N/A | N/A | 36148 | 1 | |
| 11 | Meter2 SN | RO | STR | N/A | N/A | 36200 | 16 | |
| 12 | Meter2 MFG ID | RO | STR | N/A | N/A | 36216 | 16 | |
| 13 | Meter2 TYPE | RO | STR | N/A | N/A | 36232 | 16 | |
| 14 | Meter2 Version | RO | STR | N/A | N/A | 36248 | 1 | |

**Table 3-3 Battery version definition table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 15 | BMS1 Connection Status | RO | U16 | N/A | N/A | 37002 | 1 | 0: Offline<br>1: Online |
| 16 | BMS1 Master Version | RO | U16 | N/A | N/A | 37003 | 1 | |
| 17 | BMS1 Master Type | RO | U16 | N/A | N/A | 37004 | 1 | |
| 18 | BMS1 Master SN | RO | STR | N/A | N/A | 37005 | 16 | |
| 19 | BMS1 Slave Number | RO | U16 | N/A | 1 | 37032 | 1 | Range:[0, 32], and 0 is invalid. |
| 20 | BMS1 Slave 1 Version | RO | U16 | N/A | N/A | 37033 | 1 | Maximum support for 32 channels of slave control. The number of channels actually read by the host computer is in accordance with the number of channels defined in "BMS1 Number of Slaves". The corresponding version of each battery: 37033 + (n - 1), where n is the number of slaves in the range of [1, 32]. |
| 21 | BMS1 Slave 2 Version | RO | U16 | N/A | N/A | 37034 | 1 | |
| 22 | BMS1 Slave1 SN | RO | STR | N/A | N/A | 37097 | 16 | Maximum support for 32 channels of slave control. The number of channels actually read by the host computer is in accordance with the number of channels defined in "BMS1 Number of |
| 23 | BMS1 Slave2 SN | RO | STR | N/A | N/A | 37113 | 16 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Slaves". The corresponding serial number of each battery: 37097 + 16 * (n - 1), where n is the number of slaves in the range of [1, 32]. |
| 24 | BMS1 Voltage | RO | U16 | V | 10 | 37609 | 1 | |
| 25 | BMS1 Current | RO | I16 | A | 10 | 37610 | 1 | >0, charging<br><0, discharging |
| 26 | BMS1 Ambient Temperature | RO | I16 | ℃ | 10 | 37611 | 1 | |
| 27 | BMS1 SoC | RO | U16 | % | 1 | 37612 | 1 | |
| 28 | BMS1 Max Temperature | RO | I16 | ℃ | 10 | 37617 | 1 | |
| 29 | BMS1 Min Temperature | RO | I16 | ℃ | 10 | 37618 | 1 | |
| 30 | BMS1 Max Cell Voltage | RO | U16 | mV | 1 | 37619 | 1 | |
| 31 | BMS1 Min Cell Voltage | RO | U16 | mV | 1 | 37620 | 1 | |
| 32 | BMS1 SOH | RO | U16 | % | 1 | 37624 | 1 | |
| 33 | BMS1 Fault1 | RO | Bitfield 16 | N/A | N/A | 37626 | 1 | |
| 34 | BMS1 Fault2 | RO | Bitfield 16 | N/A | N/A | 37627 | 1 | |
| 35 | BMS1 Fault3 | RO | Bitfield 16 | N/A | N/A | 37628 | 1 | |
| 36 | BMS1 Fault4 | RO | Bitfield 16 | N/A | N/A | 37629 | 1 | |
| 37 | BMS1 Fault5 | RO | Bitfield 16 | N/A | N/A | 37630 | 1 | |
| 38 | BMS1 Fault6 | RO | Bitfield 16 | N/A | N/A | 37631 | 1 | |
| 39 | BMS1 Remain Energy | RO | U16 | Wh | 0.1 | 37632 | 1 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 40 | BMS1 FCC Capacity | RO | U16 | Ah | 10 | 37633 | 1 | |
| 41 | reserve | RO | U16 | N/A | N/A | 37634 | 1 | |
| 42 | BMS1 Design Energy | RO | U16 | Wh | 0.1 | 37635 | 1 | |
| 43 | BMS1 Force to Change battery Flag | RO | U16 | N/A | N/A | 37636 | 1 | Range: [0, 1]<br>0: Reset<br>1: Set<br>Note: If set, pcs should charge the battery until the status is reset. |
| 44 | BMS2 Connection Status | RO | U16 | N/A | N/A | 37700 | 1 | 0: Offline<br>1: Online |
| 45 | BMS2 Master Version | RO | U16 | N/A | N/A | 37701 | 1 | |
| 46 | BMS2 Master Type | RO | U16 | N/A | N/A | 37702 | 1 | |
| 47 | BMS2 Master SN | RO | STR | N/A | N/A | 37703 | 16 | |
| 48 | BMS2 Slave Number | RO | U16 | N/A | 1 | 37730 | 1 | Range:[0, 32], and 0 is invalid. |
| 49 | BMS2 Slave 1 Version | RO | U16 | N/A | N/A | 37731 | 1 | Maximum support for 32 channels of slave control. The number of channels actually read by the host computer is in accordance with the number of channels defined in "BMS2 Number of Slaves". The corresponding version of each battery: 37731+ (n - 1), where n is the number of slaves in the range of |
| 50 | BMS2 Slave 2 Version | RO | U16 | N/A | N/A | 37732 | 1 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | [1, 32]. |
| 51 | BMS2 Slave1 SN | RO | STR | N/A | N/A | 37795 | 16 | Maximum support for 32 channels of slave control. The number of channels actually read by the host computer is in accordance with the number of channels defined in "BMS2 Number of Slaves". The corresponding serial number of each battery: 37795 + 16 * (n - 1), where n is the number of slaves in the range of [1, 32]. |
| 52 | BMS2 Slave2 SN | RO | STR | N/A | N/A | 37811 | 16 | |
| 53 | BMS2 Voltage | RO | U16 | V | 10 | 38307 | 1 | |
| 54 | BMS2 Current | RO | I16 | A | 10 | 38308 | 1 | >0, charging<br><0, discharging |
| 55 | BMS2 Ambient Temperature | RO | I16 | ℃ | 10 | 38309 | 1 | |
| 56 | BMS2 SoC | RO | U16 | % | 1 | 38310 | 1 | |
| 57 | BMS2 Max Temperature | RO | I16 | ℃ | 10 | 38315 | 1 | |
| 58 | BMS2 Min Temperature | RO | I16 | ℃ | 10 | 38316 | 1 | |
| 59 | BMS2 Max Cell Voltage | RO | U16 | mV | 1 | 38317 | 1 | |
| 60 | BMS2 Min Cell Voltage | RO | U16 | mV | 1 | 38318 | 1 | |
| 61 | BMS2 SOH | RO | U16 | % | 1 | 38322 | 1 | |
| 62 | BMS2 Fault1 | RO | Bitfield 16 | N/A | N/A | 38324 | 1 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 63 | BMS2 Fault2 | RO | Bitfield 16 | N/A | N/A | 38325 | 1 | |
| 64 | BMS2 Fault3 | RO | Bitfield 16 | N/A | N/A | 38326 | 1 | |
| 65 | BMS2 Fault4 | RO | Bitfield 16 | N/A | N/A | 38327 | 1 | |
| 66 | BMS2 Fault5 | RO | Bitfield 16 | N/A | N/A | 38328 | 1 | |
| 67 | BMS2 Fault6 | RO | Bitfield 16 | N/A | N/A | 38329 | 1 | |
| 68 | BMS2 Remain Energy | RO | U16 | Wh | 0.1 | 38330 | 1 | |
| 69 | BMS2 FCC Capacity | RO | U16 | Ah | 10 | 38331 | 1 | |
| 70 | reserve | RO | U16 | N/A | N/A | 38332 | 1 | |
| 71 | BMS2 Design Energy | RO | U16 | Wh | 0.1 | 38333 | 1 | |
| 72 | BMS2 Force to Change battery Flag | RO | U16 | N/A | N/A | 38334 | 1 | Range: [0, 1]<br>0: Reset<br>1: Set<br>Note: If set, pcs should charge the battery until the status is reset. |

**Table 3-4 Meter information definition table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 73 | Meter1/CT1 Connection State | RO | U16 | N/A | N/A | 38801 | 1 | 0 : Disconnect<br>1 : Connect |
| 74 | Meter1/CT1 R Phase Voltage | RO | I32 | V | 10 | 38802 | 2 | |
| 75 | Meter1/CT1 S Phase Voltage | RO | I32 | V | 10 | 38804 | 2 | |
| 76 | Meter1/CT1 T Phase Voltage | RO | I32 | V | 10 | 38806 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 77 | Meter1/CT1 R Phase Current | RO | I32 | A | 1000 | 38808 | 2 | |
| 78 | Meter1/CT1 S Phase Current | RO | I32 | A | 1000 | 38810 | 2 | |
| 79 | Meter1/CT1 T Phase Current | RO | I32 | A | 1000 | 38812 | 2 | |
| 80 | Meter1/CT1 Combined Active Power | RO | I32 | W | 10 | 38814 | 2 | |
| 81 | Meter1/CT1 R Phase Active Power | RO | I32 | W | 10 | 38816 | 2 | |
| 82 | Meter1/CT1 S Phase Active Power | RO | I32 | W | 10 | 38818 | 2 | |
| 83 | Meter1/CT1 T Phase Active Power | RO | I32 | W | 10 | 38820 | 2 | |
| 84 | Meter1/CT1 Combined Reactive Power | RO | I32 | Var | 10 | 38822 | 2 | |
| 85 | Meter1/CT1 R Phase Reactive Power | RO | I32 | Var | 10 | 38824 | 2 | |
| 86 | Meter1/CT1 S Phase Reactive Power | RO | I32 | Var | 10 | 38826 | 2 | |
| 87 | Meter1/CT1 T Phase Reactive Power | RO | I32 | Var | 10 | 38828 | 2 | |
| 88 | Meter1/CT1 Combined Apparent Power | RO | I32 | VA | 10 | 38830 | 2 | |
| 89 | Meter1/CT1 R Phase Apparent Power | RO | I32 | VA | 10 | 38832 | 2 | |
| 90 | Meter1/CT1 S Phase Apparent | RO | I32 | VA | 10 | 38834 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Power | | | | | | | |
| 91 | Meter1/CT1 T Phase Apparent Power | RO | I32 | VA | 10 | 38836 | 2 | |
| 92 | Meter1/CT1 Combined Power Factor | RO | I32 | N/A | 1000 | 38838 | 2 | |
| 93 | Meter1/CT1 R Phase Power Factor | RO | I32 | N/A | 1000 | 38840 | 2 | |
| 94 | Meter1/CT1 S Phase Power Factor | RO | I32 | N/A | 1000 | 38842 | 2 | |
| 95 | Meter1/CT1 T Phase Power Factor | RO | I32 | N/A | 1000 | 38844 | 2 | |
| 96 | Meter1/CT1 Freq | RO | I32 | Hz | 100 | 38846 | 2 | |
| 97 | Meter2/CT2 Connection State | RO | U16 | N/A | N/A | 38901 | 1 | 0 : Disconnect<br>1 : Connect |
| 98 | Meter2/CT2 R Phase Voltage | RO | I32 | V | 10 | 38902 | 2 | |
| 99 | Meter2/CT2 S Phase Voltage | RO | I32 | V | 10 | 38904 | 2 | |
| 100 | Meter2/CT2 T Phase Voltage | RO | I32 | V | 10 | 38906 | 2 | |
| 101 | Meter2/CT2 R Phase Current | RO | I32 | A | 1000 | 38908 | 2 | |
| 102 | Meter2/CT2 S Phase Current | RO | I32 | A | 1000 | 38910 | 2 | |
| 103 | Meter2/CT2 T Phase Current | RO | I32 | A | 1000 | 38912 | 2 | |
| 104 | Meter2/CT2 Combined Active Power | RO | I32 | W | 10 | 38914 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 105 | Meter2/CT2 R Phase Active Power | RO | I32 | W | 10 | 38916 | 2 | |
| 106 | Meter2/CT2 S Phase Active Power | RO | I32 | W | 10 | 38918 | 2 | |
| 107 | Meter2/CT2 T Phase Active Power | RO | I32 | W | 10 | 38920 | 2 | |
| 108 | Meter2/CT2 Combined Reactive Power | RO | I32 | Var | 10 | 38922 | 2 | |
| 109 | Meter2/CT2 R Phase Reactive Power | RO | I32 | Var | 10 | 38924 | 2 | |
| 110 | Meter2/CT2 S Phase Reactive Power | RO | I32 | Var | 10 | 38926 | 2 | |
| 111 | Meter2/CT2 T Phase Reactive Power | RO | I32 | Var | 10 | 38928 | 2 | |
| 112 | Meter2/CT2 Combined Apparent Power | RO | I32 | VA | 10 | 38930 | 2 | |
| 113 | Meter2/CT2 R Phase Apparent Power | RO | I32 | VA | 10 | 38932 | 2 | |
| 114 | Meter2/CT2 S Phase Apparent Power | RO | I32 | VA | 10 | 38934 | 2 | |
| 115 | Meter2/CT2 T Phase Apparent Power | RO | I32 | VA | 10 | 38936 | 2 | |
| 116 | Meter2/CT2 Combined Power Factor | RO | I32 | N/A | 1000 | 38938 | 2 | |
| 117 | Meter2/CT2 R Phase Power Factor | RO | I32 | N/A | 1000 | 38940 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 118 | Meter2/CT2 S Phase Power Factor | RO | I32 | N/A | 1000 | 38942 | 2 | |
| 119 | Meter2/CT2 T Phase Power Factor | RO | I32 | N/A | 1000 | 38944 | 2 | |
| 120 | Meter2/CT2 Freq | RO | I32 | Hz | 100 | 38946 | 2 | |

**Table 3-5 Basic information table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 121 | Protocol Version | RO | U32 | N/A | N/A | 39000 | 2 | For example, 0x01020304 indicates version V1.02.03.04, and the initial version is V1.01.00.00 |
| 122 | Model name | RO | STR | N/A | 1 | 39002 | 16 | |
| 123 | SN | RO | STR | N/A | 1 | 39018 | 16 | |
| 124 | PN | RO | STR | N/A | 1 | 39034 | 16 | |
| 125 | Model ID | RO | U16 | N/A | 1 | 39050 | 1 | |
| 126 | Number of strings | RO | U16 | N/A | 1 | 39051 | 1 | |
| 127 | Number of MPPTs | RO | U16 | N/A | 1 | 39052 | 1 | |
| 128 | Rated power (Pn) | RO | I32 | kW | 1000 | 39053 | 2 | |
| 129 | Maximum active power (Pmax) | RO | I32 | kW | 1000 | 39055 | 2 | |
| 130 | Maximum apparent value (Smax) | RO | I32 | kVA | 1000 | 39057 | 2 | |
| 131 | Maximum reactive power | RO | I32 | kVar | 1000 | 39059 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | (Qmax, fed into the grid) | | | | | | | |
| 132 | Maximum reactive power (Qmax, absorbed from the grid) | RO | I32 | kVar | 1000 | 39061 | 2 | |
| 133 | Status 1 | RO | Bitfield16 | N/A | 1 | 39063 | 1 | Bit0: Standby<br>Bit1: Reserved<br>Bit2: Operation<br>Bit3: Reserved<br>Bit4: Reserved<br>Bit5: Reserved<br>Bit6: Fault<br>Bit7: Reserved |
| 134 | reserve | RO | Bitfield16 | | | 39064 | 1 | |
| 135 | Status 3 | RO | Bitfield32 | N/A | 1 | 39065 | 2 | Bit0: Off-grid or not<br>0: Not off-grid<br>1: Off-grid |
| 136 | Alarm 1 | RO | Bitfield16 | N/A | 1 | 39067 | 1 | Refer to 4.1 Alarm information |
| 137 | Alarm 2 | RO | Bitfield16 | N/A | 1 | 39068 | 1 | Refer to 4.1 Alarm information |
| 138 | Alarm 3 | RO | Bitfield16 | N/A | 1 | 39069 | 1 | Refer to 4.1 Alarm information |
| 139 | PV1 voltage | RO | I16 | V | 10 | 39070 | 1 | Supports up to 24 channels of string information. The actual number of reading channels of the host computer is based on the number of channels defined by "number of strings". The actual |
| 140 | PV1 current | RO | I16 | A | 100 | 39071 | 1 | |
| 141 | PV2 voltage | RO | I16 | V | 10 | 39072 | 1 | |
| 142 | PV2 current | RO | I16 | A | 100 | 39073 | 1 | |
| 143 | PV3 voltage | RO | I16 | V | 10 | 39074 | 1 | |
| 144 | PV3 current | RO | I16 | A | 100 | 39075 | 1 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 145 | PV4 voltage | RO | I16 | V | 10 | 39076 | 1 | corresponding voltage and current register addresses of each string are<br>PVn voltage: 39070 + 2*(n - 1),<br>PVn current: 39071 + 2*(n – 1).<br>Where n is the string number range is [1, 24] |
| 146 | PV4 current | RO | I16 | A | 100 | 39077 | 1 | |
| 147 | Total PV input power | RO | I32 | kW | 1000 | 39118 | 2 | - |
| 148 | reserve | RO | U16 | | | 39120 | 1 | - |
| 149 | reserve | RO | U16 | | | 39121 | 1 | - |
| 150 | reserve | RO | U16 | | | 39122 | 1 | - |
| 151 | Grid R Phase Voltage | RO | I16 | V | 10 | 39123 | 1 | |
| 152 | Grid S Phase Voltage | RO | I16 | V | 10 | 39124 | 1 | |
| 153 | Grid T Phase Voltage | RO | I16 | V | 10 | 39125 | 1 | |
| 154 | Inverter R Phase Current | RO | I32 | A | 1000 | 39126 | 2 | |
| 155 | Inverter S Phase Current | RO | I32 | A | 1000 | 39128 | 2 | |
| 156 | Inverter T Phase Current | RO | I32 | A | 1000 | 39130 | 2 | |
| 157 | reserve | RO | U32 | | | 39132 | 2 | |
| 158 | Active power | RO | I32 | kW | 1000 | 39134 | 2 | |
| 159 | Reactive power | RO | I32 | kVar | 1000 | 39136 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 160 | Power Factor | RO | I16 | N/A | 1000 | 39138 | 1 | |
| 161 | Grid Frequency | RO | I16 | Hz | 100 | 39139 | 1 | |
| 162 | reserve | RO | U16 | | | 39140 | 1 | |
| 163 | Internal temperature | RO | I16 | ℃ | 10 | 39141 | 1 | |
| 164 | reserve | RO | U16 | | | 39142 | 1 | |
| 165 | reserve | RO | U16 | | | 39143 | 1 | |
| 166 | reserve | RO | U16 | | | 39144 | 1 | |
| 167 | reserve | RO | U32 | | | 39145 | 2 | |
| 168 | reserve | RO | U32 | | | 39147 | 2 | |
| 169 | Cumulative power generation | RO | U32 | kWh | 100 | 39149 | 2 | |
| 170 | Power generation on the day | RO | U32 | kWh | 100 | 39151 | 2 | |
| 171 | reserve | RO | U16 | | | 39153 | 1 | |
| 172 | reserve | RO | U32 | | | 39154 | 2 | |
| 173 | reserve | RO | U16 | | | 39156 | 1 | |
| 174 | reserve | RO | U16 | | | 39157 | 1 | |
| 175 | reserve | RO | U32 | | | 39158 | 2 | |
| 176 | reserve | RO | U16 | | | 39160 | 1 | |
| 177 | reserve | RO | U16 | | | 39161 | 1 | |
| 178 | [Energy storage module 1] Charge and discharge | RO | I32 | W | 1 | 39162 | 2 | > 0: charging<br>< 0: discharging |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | power * | | | | | | | |
| 179 | reserve | RO | U32 | | | 39164 | 2 | |
| 180 | reserve | RO | U32 | | | 39166 | 2 | |
| 181 | [Meter collection] Active power * | RO | I32 | W | 1 | 39168 | 2 | > 0: Feed power to the grid<br>< 0: Take power from the grid |
| 182 | reserve | RO | U16 | | | 39170 | 1 | |
| 183 | reserve | RO | U16 | | | 39171 | 1 | |
| 184 | reserve | RO | U16 | | | 39172 | 1 | |
| 185 | reserve | RO | U16 | | | 39200 | 1 | |
| 186 | EPS R Phase Voltage | RO | U16 | V | 10 | 39201 | 1 | |
| 187 | EPS S Phase Voltage | RO | U16 | V | 10 | 39202 | 1 | |
| 188 | EPS T Phase Voltage | RO | U16 | V | 10 | 39203 | 1 | |
| 189 | EPS R Phase Current | RO | I32 | A | 1000 | 39204 | 2 | |
| 190 | EPS S Phase Current | RO | I32 | A | 1000 | 39206 | 2 | |
| 191 | EPS T Phase Current | RO | I32 | A | 1000 | 39208 | 2 | |
| 192 | EPS R Phase Power | RO | I32 | W | 1 | 39210 | 2 | |
| 193 | EPS S Phase Power | RO | I32 | W | 1 | 39212 | 2 | |
| 194 | EPS T Phase Power | RO | I32 | W | 1 | 39214 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 195 | EPS Combined Power | RO | I32 | W | 1 | 39216 | 2 | |
| 196 | EPS Frequency | RO | I16 | Hz | 100 | 39218 | 1 | |
| 197 | Load R Phase Power | RO | I32 | W | 1 | 39219 | 2 | |
| 198 | Load S Phase Power | RO | I32 | W | 1 | 39221 | 2 | |
| 199 | Load T Phase Power | RO | I32 | W | 1 | 39223 | 2 | |
| 200 | Load Combined Power | RO | I32 | W | 1 | 39225 | 2 | |
| 201 | Battery1 Voltage | RO | I16 | V | 10 | 39227 | 1 | |
| 202 | Battery1 Current | RO | I32 | A | 1000 | 39228 | 2 | >0, Battery discharge<br><0, Battery charge |
| 203 | Battery1 Power | RO | I32 | W | 1 | 39230 | 2 | >0, Battery discharge<br><0, Battery charge |
| 204 | Battery2 Voltage | RO | I16 | V | 10 | 39232 | 1 | |
| 205 | Battery2 Current | RO | I32 | A | 1000 | 39233 | 2 | >0, Battery discharge<br><0, Battery charge |
| 206 | Battery2 Power | RO | I32 | W | 1 | 39235 | 2 | >0, Battery discharge<br><0, Battery charge |
| 207 | Battery Combined Power | RO | I32 | W | 1 | 39237 | 2 | |
| 208 | reserve | RO | I16 | | | 39239 | 1 | |
| 209 | reserve | RO | I16 | | | 39240 | 1 | |
| 210 | reserve | RO | I16 | | | 39241 | 1 | |
| 211 | reserve | RO | I32 | | | 39242 | 2 | |
| 212 | reserve | RO | I32 | | | 39244 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 213 | reserve | RO | I32 | | | 39246 | 2 | |
| 214 | INV R Phase Active Power | RO | I32 | W | 1 | 39248 | 2 | |
| 215 | INV S Phase Active Power | RO | I32 | W | 1 | 39250 | 2 | |
| 216 | INV T Phase Active Power | RO | I32 | W | 1 | 39252 | 2 | |
| 217 | reserve | RO | I32 | | | 39254 | 2 | |
| 218 | INV R Phase Reactive Power | RO | I32 | Var | 1 | 39256 | 2 | |
| 219 | INV S Phase Reactive Power | RO | I32 | Var | 1 | 39258 | 2 | |
| 220 | INV T Phase Reactive Power | RO | I32 | Var | 1 | 39260 | 2 | |
| 221 | reserve | RO | I32 | | | 39262 | 2 | |
| 222 | INV R Phase Apparent Power | RO | I32 | VA | 1 | 39264 | 2 | |
| 223 | INV S Phase Apparent Power | RO | I32 | VA | 1 | 39266 | 2 | |
| 224 | INV T Phase Apparent Power | RO | I32 | VA | 1 | 39268 | 2 | |
| 225 | INV Combined Apparent Power | RO | I32 | VA | 1 | 39270 | 2 | |
| 226 | INV Frequency R | RO | I16 | Hz | 100 | 39272 | 1 | |
| 227 | INV Frequency S | RO | I16 | Hz | 100 | 39273 | 1 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 228 | INV Frequency T | RO | I16 | Hz | 100 | 39274 | 1 | |
| 229 | Available Import Power | RO | I32 | W | 1 | 39275 | 2 | |
| 230 | Available Export Power | RO | I32 | W | 1 | 39277 | 2 | |
| 231 | PV1 Power | RO | I32 | W | 1 | 39279 | 2 | Maximum support for 24 channels of string information. The number of channels actually read by the host computer is in accordance with the number of channels defined in "Number of strings". The PV Power registers corresponding to each string are as follows<br>PVn Power: 39279 + 2 * (n - 1)<br>Where n is the string number, the range is [1, 24]. |
| 232 | PV2 Power | RO | I32 | W | 1 | 39281 | 2 | |
| 233 | PV3 Power | RO | I32 | W | 1 | 39283 | 2 | |
| 234 | PV4 Power | RO | I32 | W | 1 | 39285 | 2 | |
| 235 | MPPT1 Voltage | RO | I16 | V | 10 | 39327 | 1 | Maximum support for 24 channels of string information. The number of channels actually read by the host computer is in accordance with the number of channels defined in "Number of strings". The MPPT |
| 236 | MPPT1 Current | RO | I16 | A | 100 | 39328 | 1 | |
| 237 | MPPT1 Power | RO | I32 | W | 1 | 39329 | 2 | |
| 238 | MPPT2 Voltage | RO | I16 | V | 10 | 39331 | 1 | |
| 239 | MPPT2 Current | RO | I16 | A | 100 | 39332 | 1 | |
| 240 | MPPT2 Power | RO | I32 | W | 1 | 39333 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 241 | MPPT3 Voltage | RO | I16 | V | 10 | 39335 | 1 | Voltage and Current and Power registers corresponding to each string are as follows<br>MPPTn Volt: 39327 + 4*(n - 1)<br>MPPTn Curr: 39328 + 4*(n - 1)<br>MPPTn Power: 39329 + 4*(n - 1)<br>Where n is the string number, the range is [1, 24]. |
| 242 | MPPT3 Current | RO | I16 | A | 100 | 39336 | 1 | |
| 243 | MPPT3 Power | RO | I32 | W | 1 | 39337 | 2 | |
| 244 | System SoC | RO | U16 | % | 1 | 39423 | 1 | Range: [0, 100] |
| 245 | Power Factor for Phase-R | RO | I16 | N/A | 1000 | 39424 | 1 | |
| 246 | Power Factor for Phase-S | RO | I16 | N/A | 1000 | 39425 | 1 | |
| 247 | Power Factor for Phase-T | RO | I16 | N/A | 1000 | 39426 | 1 | |

**Table 3-6 Power and Energy data table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 248 | reserve | RO | U16 | N/A | N/A | 39600 | 1 | |
| 249 | PV total power | RO | U32 | kWh | 100 | 39601 | 2 | |
| 250 | Total PV power today | RO | U32 | kWh | 100 | 39603 | 2 | |
| 251 | Total charging capacity | RO | U32 | kWh | 100 | 39605 | 2 | |
| 252 | Today's total charging capacity | RO | U32 | kWh | 100 | 39607 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 253 | Total discharge power | RO | U32 | kWh | 100 | 39609 | 2 | |
| 254 | Today's total discharge power | RO | U32 | kWh | 100 | 39611 | 2 | |
| 255 | Total power of feeder network | RO | U32 | kWh | 100 | 39613 | 2 | |
| 256 | Today's total feeder power | RO | U32 | kWh | 100 | 39615 | 2 | |
| 257 | Total power taken | RO | U32 | kWh | 100 | 39617 | 2 | |
| 258 | Today's total electricity consumption | RO | U32 | kWh | 100 | 39619 | 2 | |
| 259 | Output total power | RO | U32 | kWh | 100 | 39621 | 2 | |
| 260 | Total power output today | RO | U32 | kWh | 100 | 39623 | 2 | |
| 261 | Enter total power | RO | U32 | kWh | 100 | 39625 | 2 | |
| 262 | Enter total power today | RO | U32 | kWh | 100 | 39627 | 2 | |
| 263 | Total load power | RO | U32 | kWh | 100 | 39629 | 2 | |
| 264 | Total load power today | RO | U32 | kWh | 100 | 39631 | 2 | |
| 265 | Total BMS charging capacity | RO | U32 | kWh | 100 | 39633 | 2 | Note: Unsupported |
| 266 | Today's total BMS charging capacity | RO | U32 | kWh | 100 | 39635 | 2 | Note: Unsupported |
| 267 | Total BMS discharge power | RO | U32 | kWh | 100 | 39637 | 2 | Note: Unsupported |
| 268 | Today's total BMS discharge power | RO | U32 | kWh | 100 | 39639 | 2 | Note: Unsupported |

**Table 3-7 Executive command-1 table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 269 | reserve | WO | U16 | N/A | N/A | 45000 | 1 | |
| 270 | reserve | WO | U16 | N/A | N/A | 45001 | 1 | |
| 271 | Factory Reset | WO | U16 | N/A | N/A | 45002 | 1 | Range: [0, 1]<br>0: Invalid; 1: Active |
| 272 | Battery power active | WO | U16 | N/A | N/A | 45003 | 1 | Range: [0, 1]<br>0:Invalid; 1: Active<br>Note: Only the H3 Smart series is supported. |
| 273 | reserve | WO | U16 | N/A | N/A | 45004 | 1 | |
| 274 | Battery power shut-dowm | WO | U16 | N/A | N/A | 45005 | 1 | Range: [0, 1]<br>0:Invalid; 1: Active<br>Note: Only the H3 Smart series is supported. |
| 275 | Battery power ON/OFF | RO | U16 | N/A | N/A | 45006 | 1 | Range: [0, 1]<br>0: OFF<br>1: ON |
| 276 | Battery Connect Enable | RW | U16 | N/A | N/A | 45007 | 1 | Range: [0, 1]<br>0: Disable<br>1: Enable<br>Note: Only the H3 Smart series is supported. |

**Table 3-8 Remote Control table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 277 | reserve | RO | U16 | N/A | N/A | 46000 | 1 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 278 | Remote Control | RW | Bitfield 16 | N/A | 1 | 46001 | 1 | Bit 0: Remote Control enable<br>0: Disable<br>1: Enable<br><br>Bit 1: Definition for positive direction<br>0: power-generation system<br>1: power-consumption system<br><br>Bit 3:2 : Controlled target<br>00: AC<br>01: Battery<br>10: Grid (CT/Meter)<br>11: AC (Grid first)<br><br>Bit 8:4 Reserved<br><br>Bit 9: PV Enable<br>0: Disable<br>1: Enable<br><br>Bit 15:10 Reserved |
| 279 | Remote Timeout Set | RW | U16 | s | 1 | 46002 | 1 | |
| 280 | Remote Control Active Power Command | RW | I32 | W | 1 | 46003 | 2 | |
| 281 | Remote Control Reactive Power Command | RW | I32 | Var | 1 | 46005 | 2 | |
| 282 | Remote Timeout Countdown | RO | U16 | s | 1 | 46007 | 1 | |
| 283 | Pwr_limit Bat_Up | RO | I32 | W | 1 | 46018 | 2 | The maximum avaliable discharge power of the battery |
| 284 | Pwr_limit Bat_Dn | RO | I32 | W | 1 | 46020 | 2 | The maximum avaliable charging power of the battery |
| 285 | reserve | | | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 286 | Remote Control PV Power Command | RW | I32 | W | 1 | 46026 | 2 | |

**Table 3-9 Peak Shaving table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 287 | reserve | RW | U16 | N/A | N/A | 46500 | 1 | |
| 288 | Import Power Limit * | RW | I32 | W | 1 | 46501 | 2 | [0, 100000] |
| 289 | Threshold SOC * | RW | U16 | % | 1 | 46503 | 1 | [10, 100] |
| 290 | Export Peak Limit * | RW | I32 | W | 1 | 46504 | 2 | [0, 100000] |
| 291 | ChrInLowImport * | RW | U16 | N/A | 1 | 46506 | 1 | 0: NO Mode<br>1: NC Mode<br>2: Timing Mode |
| 292 | ChrInLowTime1-StartHour * | RW | U16 | N/A | 1 | 46507 | 1 | [0, 23] |
| 293 | ChrInLowTime1-StartMinute * | RW | U16 | N/A | 1 | 46508 | 1 | [0, 59] |
| 294 | ChrInLowTime1-EndHour * | RW | U16 | N/A | 1 | 46509 | 1 | [0, 23] |
| 295 | ChrInLowTime1-EndMinute * | RW | U16 | N/A | 1 | 46510 | 1 | [0, 59] |
| 296 | ChrInLowTime2-StartHour * | RW | U16 | N/A | 1 | 46511 | 1 | [0, 23] |
| 297 | ChrInLowTime2-StartMinute * | RW | U16 | N/A | 1 | 46512 | 1 | [0, 59] |
| 298 | ChrInLowTime2 -EndHour * | RW | U16 | N/A | 1 | 46513 | 1 | [0, 23] |
| 299 | ChrInLowTime2-EndMinute * | RW | U16 | N/A | 1 | 46514 | 1 | [0, 59] |

**Table 3-10 Executive command-2 table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 300 | reserve | RW | U16 | N/A | N/A | 46601 | 1 | |
| 301 | reserve | RW | U16 | N/A | N/A | 46602 | 1 | |
| 302 | reserve | RW | U16 | N/A | N/A | 46603 | 1 | |
| 303 | reserve | RW | U16 | N/A | N/A | 46604 | 1 | |
| 304 | reserve | RW | U16 | N/A | N/A | 46605 | 1 | |
| 305 | reserve | RW | U16 | N/A | N/A | 46606 | 1 | |
| 306 | Battery maximum charging current * | RW | I16 | A | 10 | 46607 | 1 | H3:[0,26]<br>H3-G2:[0,50]<br>H3Pro:[0,50]<br>KH:[0,50]<br>H1:[ 0 , 40 ]<br>H1-G2:[ 0, 40]<br>812: [0, 50] |
| 307 | Battery maximum discharge current * | RW | I16 | A | 10 | 46608 | 1 | H3:[0,26]<br>H3-G2:[0,50]<br>H3Pro:[0,50]<br>KH:[0,50]<br>H1:[ 0 , 50 ]<br>H1-G2:[ 0, 40]<br>812: [0, 50] |
| 308 | Minimum SoC * | RW | U16 | % | 1 | 46609 | 1 | [10,100] |
| 309 | Maximum SoC * | RW | U16 | % | 1 | 46610 | 1 | [10,100] |
| 310 | Minimum SoC OnGrid * | RW | U16 | % | 1 | 46611 | 1 | [10,100] |
| 311 | EPS Frequency Select * _ | RW | U16 | N/A | N/A | 46612 | 1 | 0: invalid<br>1:50Hz |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | 2: 60Hz |
| 312 | EPS Output * – | RW | U16 | N/A | N/A | 46613 | 1 | 0:Disable<br>2:EPS Mode<br>3:Ups Mode |
| 313 | Balance Load * | RW | U16 | N/A | N/A | 46614 | 1 | 0:Disable<br>1:Enable |
| 314 | Balance Logic First * | RW | U16 | N/A | N/A | 46615 | 1 | 0:Disable<br>1:Enable |
| 315 | Export Power Limit | RW | I32 | W | 1 | 46616 | 2 | [0, Pmax] |
| 316 | Import Current Limit | RW | I16 | A | 10 | 46618 | 1 | H3-G2: [0, 200]<br>KH: [2, 100]<br>812: [2, 100] |
| 317 | Export Current Limit | RW | I16 | A | 10 | 46619 | 1 | H3-G2: [0, 200]<br>KH: [2, 100]<br>812: [2, 100] |
| 318 | Maximum SoC From Grid * | RW | U16 | % | 1 | 46620 | 1 | [10,100] |

**Table 3-11 Time Period table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 319 | Time Mode Flag | RW | U16 | N/A | 1 | 48000 | 1 | Time Period Mode<br>0:Disable<br>1:Enable |
| 320 | reserve | RW | U16 | N/A | 1 | 48001 | 1 | |
| 321 | reserve | RW | U16 | N/A | 1 | 48002 | 1 | |
| 322 | reserve | RW | U16 | N/A | 1 | 48003 | 1 | |
| 323 | reserve | RW | U16 | N/A | 1 | 48004 | 1 | |
| 324 | reserve | RW | U16 | N/A | 1 | 48005 | 1 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 325 | reserve | RW | U16 | N/A | 1 | 48006 | 1 | |
| 326 | reserve | RW | U16 | N/A | 1 | 48007 | 1 | |
| 327 | reserve | RW | U16 | N/A | 1 | 48008 | 1 | |
| 328 | reserve | RW | U16 | N/A | 1 | 48009 | 1 | |
| 329 | Time Group ( N ) Disable / Enable | RW | U16 | N/A | 1 | 48010 + 10*(N-1) | 1 | Time Group N<br>0:Disable<br>1:Enable<br><br>Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| 330 | Time Group ( N ) Start Time – Hour Minute | RW | U16 | N/A | 1 | 48011 + 10*(N-1) | 1 | High Byte is hour, and its range is [0, 23].<br>Low Byte is minute, and its range is [0, 59].<br><br>Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| 331 | Time Group ( N ) End Time – Hour Minute | RW | U16 | N/A | 1 | 48012 + 10*(N-1) | 1 | High Byte is hour, and its range is [0, 23].<br>Low Byte is minute, and its range is [0, 59].<br><br>Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 332 | Time Group ( N ) Work Mode | RW | U16 | N/A | 1 | 48013 + 10*(N-1) | 1 | 1: Self-Use<br>2: Feed-In<br>~~3: BackUp~~<br>~~4: Peak Shaving~~<br>~~5: Power Station~~<br>6: Force Charge<br>7: Force Discharge<br><br>Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| 333 | Time Group ( N ) MaxSoC_FromGrid | RW | U8 | % | 1 | 48014 + 10*(N-1) | 1 | Range: [10, 100]<br><br>Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| | Time Group ( N ) MinSoC OnGrid | RW | U8 | % | 1 | | | Range: [10, 100]<br>Where, >= MinSoC && <= MaxSoC_FromGrid<br><br>Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| 334 | Time Group ( N ) FC/FDSOC | RW | U16 | % | 1 | 48015 + 10*(N-1) | 1 | FCSOC Range: [10, MaxSoC_FromGrid]<br><br>FDSOC Range: [MinSoC OnGrid, 100] |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| 335 | Time Group ( N ) FC/FDPWR | RW | U16 | W | 1 | 48016 + 10*(N-1) | 1 | Range: [0, Pmax]<br><br>Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| 336 | Time Group ( N ) reserve | RW | U16 | N/A | 1 | 48017 + 10*(N-1) | 1 | Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| 337 | Time Group ( N ) reserve | RW | U16 | N/A | 1 | 48018 + 10*(N-1) | 1 | Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |
| 338 | Time Group ( N ) reserve | RW | U16 | N/A | 1 | 48019 + 10*(N-1) | 1 | Note: A maximum of 96 time groups are supported, and its range of value N is [1, 96]. |

Note:

- Time period table is divided into 97 blocks of contiguous registers, namely blocks of contiguous registers 48000 to 48009, 48010 to 48019, 48020 to 48029, …, 48960 to 48969. The system supports each block of contiguous registers can be read and written simultaneously.
- Or, time period table is divided into 13 blocks of contiguous registers, namely blocks of contiguous registers 48000 to 48009, 48010 to 48089, 48090 to 48169, 48170 to 48249, 48250 to 48329, 48330 to 48409, 48410 to 48489, 48490 to 48569, 48570 to 48649, 48650 to 48729, 48730 to 48809, 48810 to 48889, 48890 to 48969. The system supports each block of contiguous registers can be read and written simultaneously.

**Table 3-12 Executive command-3 table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 339 | System Time | RW | U32 | N/A | N/A | 49000 | 2 | [946684800,3155759999] local time |
| 340 | reserve | RW | U16 | | | 49002 | 1 | |
| 341 | reserve | RW | U16 | | | 49003 | 1 | |
| 342 | reserve | RW | U16 | | | 49004 | 1 | |
| 343 | [ Power Grid Scheduling ] None Power compensation (PF) | RW | I16 | N/A | 1000 | 49005 | 1 | (-1, -0.8] U [0.8, 1] |
| 344 | [Power Grid Scheduling] None Power compensation (Q/S) | RW | I16 | N/A | 1000 | 49006 | 1 | [-1.000,+1.000] The equipment end converts this value into a fixed value of Q for reactive power control; where S is Smax |
| 345 | [Power grid dispatch] Active power percentage derating (0.1%) | RW | I16 | % | 10 | 49007 | 1 | Range: [0, 100.0 ] Active power fine adjustment interface |
| 346 | [Power grid dispatch] Fixed Active power derated (W) | RW | U32 | W | 1 | 49008 | 2 | Range: [0, Pmax] |
| 347 | [Power grid dispatch] Night Reactive power compensation (kVar) | RW | I32 | kVar | 1000 | 49010 | 2 | Range: [-Qmax, Qmax] |
| 348 | reserve | RW | MLD | | | 49012 | 21 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 349 | reserve | RW | MLD | | | 49033 | 21 | |
| 350 | reserve | RW | MLD | | | 49054 | 21 | |
| 351 | reserve | RW | U16 | | | 49075 | 1 | |
| 352 | reserve | RW | U16 | | | 49076 | 1 | |
| 353 | Power on | RW | U16 | N/A | N/A | 49077 | 1 | Range: [0,1]<br>0: invalid 1: valid<br><br>Note: To get the on/off status, query the register 49228. |
| 354 | Shut down | RW | U16 | N/A | N/A | 49078 | 1 | Range: [0,1]<br>0: invalid 1: valid<br><br>Note: To get the on/off status, query the register 49228. |
| 355 | Grid standard code | RW | U16 | N/A | N/A | 49079 | 1 | Reference 4.2 Power Grid Standard Code |
| 356 | reserve | RW | U32 | | | 49080 | 2 | |
| 357 | reserve | RW | U32 | | | 49082 | 2 | |
| 358 | reserve | RW | U32 | | | 49084 | 2 | |
| 359 | reserve | RW | U16 | | | 49086 | 1 | |
| 360 | reserve | RW | U16 | | | 49087 | 1 | |
| 361 | reserve | RW | U16 | | | 49088 | 1 | |
| 362 | reserve | RW | MLD | | | 49089 | 41 | |
| 363 | reserve | RW | U32 | | | 49130 | 2 | |
| 364 | reserve | RW | U32 | | | 49132 | 2 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 365 | reserve | RW | U32 | | | 49134 | 2 | |
| 366 | Grid point power limit | RW | I32 | W | 1 | 49136 | 2 | [0, Pmax]<br>Default value: Pmax |
| 367 | reserve | RW | U16 | | | 49138 | 1 | |
| 368 | reserve | RW | U16 | | | 49139 | 1 | |
| 369 | reserve | RW | U16 | | | 49140 | 1 | |
| 370 | reserve | RW | U32 | | | 49141 | 2 | |
| 371 | reserve | RW | MLD | | | 49143 | 41 | |
| 372 | Work mode | RW | U16 | N/A | N/A | 49203 | 1 | 1:Self Use<br>2:Feedin Priority<br>3:BackUp<br>4:Peak Shaving<br>~~6:Force Charge~~<br>~~7:Force Discharge~~ |
| 373 | DRM | RW | U16 | N / A | N/A | 49206 | 1 | 0: Disable<br>1: Enable<br>(Only valid under Australian safety regulations) |
| 374 | Meter1 /CT1 | RW | U16 | N / A | N/A | 49207 | 1 | 0: OFF<br>1: Meter 1-PHASE<br>2:CT<br>3:Meter 3-PHASE |
| 375 | Meter2 /CT2 * | RW | U16 | N / A | N/A | 49208 | 1 | 0: OFF<br>1: Meter 1-PHASE<br>2: CT<br>3: Meter 3-PHASE |
| 376 | BUZZER | RW | U16 | N / A | N/A | 49209 | 1 | 0: Disable<br>1: Enable |
| 377 | MPPT Switch (PV | RW | U16 | N / A | N/A | 49210 | 1 | 0: Disable<br>1: Enable |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Shutdown) | | | | | | | |
| 378 | Relay1 Switch | RW | U16 | N/A | N/A | 49211 | 1 | 0:Disable<br>1:Enable |
| 379 | Relay2 Switch | RW | U16 | N/A | N/A | 49212 | 1 | 0:Disable<br>1:Enable |
| 380 | Brightness Level | RW | U16 | % | 1 | 49221 | 1 | Range: [0,100] |
| 381 | Date-Year | RW | U16 | N/A | 1 | 49222 | 1 | Range: [2000, 2099] |
| 382 | Date-Month | RW | U16 | N/A | 1 | 49223 | 1 | Range: [1, 12] |
| 383 | Date-Day | RW | U16 | N/A | 1 | 49224 | 1 | Range: [1, 31] |
| 384 | Time-Hour | RW | U16 | N/A | 1 | 49225 | 1 | Range: [0, 23] |
| 385 | Time-Minute | RW | U16 | N/A | 1 | 49226 | 1 | Range: [0, 59] |
| 386 | Time-Second | RW | U16 | N/A | 1 | 49227 | 1 | Range: [0, 59] |
| 387 | System Power State | RO | U16 | N/A | 1 | 49228 | 1 | 0: Turn OFF<br>1: Turn ON |
| 388 | Idle State | RW | U16 | N/A | 1 | 49229 | 1 | 0: Disable<br>1: Enable |
| 389 | Idle Loadpower Threshold | RW | U16 | W | 1 | 49230 | 1 | Range:<br>H3: 100~200<br>H3Pro: 100~600 |
| 390 | Clear Idle Count | WO | U16 | N/A | 1 | 49231 | 1 | 0: Clear Idle Count |
| 391 | Key Password | RW | STR | N/A | 1 | 49232 | 8 | |
| 392 | Network Status | RO | U16 | N/A | 1 | 49240 | 1 | 0: Not connected<br>1: Disconnection<br>2: Connection |
| 393 | Ripple Control Enable | RW | U16 | N/A | 1 | 49241 | 1 | 0: Disable<br>1: Enable |
| 394 | Trigger Signal | RO | U16 | N/A | 1 | 49242 | 1 | Bit0: K1 status<br>Bit1: K2 status<br>Bit2: K3 status<br>Bit3: K4 status |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 395 | K1 Power Ratio | RW | U16 | % | 1 | 49243 | 1 | Range: [0, 100] |
| 396 | K2 Power Ratio | RW | U16 | % | 1 | 49244 | 1 | Range: [0, 100] |
| 397 | K3 Power Ratio | RW | U16 | % | 1 | 49245 | 1 | Range: [0, 100] |
| 398 | K4 Power | RW | U16 | % | 1 | 49246 | 1 | Reserved<br>Range: [0, 100] |
| 399 | OffGridMaster | RW | U16 | N/A | 1 | 49247 | 1 | 0: Free Mode<br>1: Master Mode |
| 400 | Meter Compensation | RW | I16 | W | 1 | 49248 | 1 | Range: [-500, 500] |
| 401 | GFCI Current | RO | I16 | A | 100 | 49249 | 1 | |
| 402 | PV Power Limited Flag | RO | U16 | N/A | 1 | 49251 | 1 | 0: Invalid<br>1: Active |
| 403 | G100 State | RW | U16 | N/A | 1 | 49252 | 1 | 0: Disbale<br>1: State1<br>2: State2<br>3: State3 |
| 404 | G100 Reset For End User | WO | U16 | N/A | 1 | 49253 | 1 | 0: Invalid<br>1: Active |
| 405 | G100 Reset For Admin | WO | U16 | N/A | 1 | 49254 | 1 | 0: Invalid<br>1: Active |
| 406 | Grid Charge Flag | RW | U16 | N/A | 1 | 49255 | 1 | 0: Disable<br>1: Enable<br>Note: MQ Only |
| 407 | ECO Mode | RW | U16 | N/A | 1 | 49256 | 1 | 0: Disable<br>1: Enable<br>Note: MQ Only |
| 408 | Easy Mode Power | RW | I32 | W | 1 | 49257 | 2 | Range: [0, 65535]<br>Note: MQ Only |
| 409 | Low Power Mode | RW | U16 | N/A | 1 | 49259 | 1 | 0: Disable<br>1: Enable<br>Note: MQ Only |
| 410 | reserve | RW | U16 | N/A | | | | |
| 411 | PV Limited Power | RW | I32 | W | 1 | 49501 | 2 | Range: [0, 300000] |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 412 | Forced Standby | RW | U16 | N/A | 1 | 49503 | 1 | 0: Disable<br>1: Forced Cold Standby<br>2: Forced Hot Standby |
| 413 | Shadow Scan (Global MPPT) | RW | U16 | N/A | 1 | 49504 | 1 | 0: Disable<br>1: Enable |
| 414 | Shadow Scan Interval (Global MPPT) | RW | U16 | Min | 1 | 49505 | 1 | Range: [30, 360]<br>Default as 120min |
| 415 | reserve | | | | | | | |

**Table 3-13 Battery warm up table**

| Serial number | Signal name | Read and write | Type | Unit | Gain | Address | Number | Scope |
|---|---|---|---|---|---|---|---|---|
| 416 | BMS Warm Up Enable | RW | U16 | N/A | 1 | 53400 | 1 | 0: Disable<br>1: Enable |
| 417 | Start Temperature | RW | I16 | ℃ | 1 | 53401 | 1 | It set as register 53412. |
| 418 | Stop Temperature | RW | I16 | ℃ | 1 | 53402 | 1 | It set as register 53413. |
| 419 | Time1 Enable | RW | U16 | N/A | 1 | 53403 | 1 | 0: Disable<br>1: Enable |
| 420 | Time1 Start-Hour Minute | RW | U16 | N/A | 1 | 53404 | 1 | High Byte is hour, and its range is [0, 23].<br>Low Byte is minute, and its range is [0, 59]. |
| 421 | Time1 End-Hour Minute | RW | U16 | N/A | 1 | 53405 | 1 | High Byte is hour, and its range is [0, 23].<br>Low Byte is minute, and its range is [0, 59]. |
| 422 | Time2 Enable | RW | U16 | N/A | 1 | 53406 | 1 | 0: Disable<br>1: Enable |
| 423 | Time2 Start-Hour Minute | RW | U16 | N/A | 1 | 53407 | 1 | High Byte is hour, and its range is [0, 23]. |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Low Byte is minute, and its range is [0, 59]. |
| 424 | Time2 End-Hour Minute | RW | U16 | N/A | 1 | 53408 | 1 | High Byte is hour, and its range is [0, 23].<br>Low Byte is minute, and its range is [0, 59]. |
| 425 | Time3 Enable | RW | U16 | N/A | 1 | 53409 | 1 | 0: Disable<br>1: Enable |
| 426 | Time3 Start-Hour Minute | RW | U16 | N/A | 1 | 53410 | 1 | High Byte is hour, and its range is [0, 23].<br>Low Byte is minute, and its range is [0, 59]. |
| 427 | Time3 End-Hour Minute | RW | U16 | N/A | 1 | 53411 | 1 | High Byte is hour, and its range is [0, 23].<br>Low Byte is minute, and its range is [0, 59]. |
| 428 | BMS Start Temperature Range | RO | I16 | ℃ | 1 | 53412 | 1 | High Byte is minimum start temperature;<br>Low Byte is maximum start temperature. |
| 429 | BMS Stop Temperature Range | RO | I16 | ℃ | 1 | 53413 | 1 | High Byte is minimum stop temperature;<br>Low Byte is maximum stop temperature. |
| 430 | BMS-1 Warm Up Status | RO | U16 | N/A | 1 | 53414 | 1 | High Byte is BMS-1 warm up state:<br>0: The battery is in a stopped warm up state;<br>1: The battery is in a ready warm up state;<br>2: The battery is in the warm up state;<br>3: The battery is in a ready stopped warm up state;<br>4: Battery is in self |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | warm up state;<br><br>Low Byte is Battery warm up function enable flag:<br>0: The battery has no warm up function;<br>1: The battery has a warm up function; |
| 431 | BMS-2 Warm Up Status | RO | U16 | N/A | 1 | 53415 | 1 | High Byte is BMS-2 warm up state:<br>0: The battery is in a stopped warm up state;<br>1: The battery is in a ready warm up state;<br>2: The battery is in the warm up state;<br>3: The battery is in a ready stopped warm up state;<br>4: Battery is in self warm up state;<br><br>Low Byte is Battery warm up function enable flag:<br>0: The battery has no warm up function;<br>1: The battery has a warm up function; |

Note:

- Single BMS system: Please read the values of registers 53412 to 53414 to obtain the warm-up status, start temperature, and stop temperature of the battery system first, and these three registers provide information about the heating status of the battery system as well as the start and stop temperature ranges. Then, you can write to multiple registers 53400 to 53411 to control the heating strategy of the battery system.
- Dual BMS system: Please read the values of registers 53412 to 53415 to obtain the warm-up status, start temperature, and stop temperature of the battery system first, and these four registers provide information

about the heating status of the battery system as well as the start and stop temperature ranges. Then, you can writes to multiple registers 53400 to 53411 to control the heating strategy of the battery system.

**Notice**

marked * are only supported by some models or standard codes.

# 4 Interface usage instructions

## 4.1 Alarm information

**Table 4-1** Alarm information

| Number | Alarm | Bit | Alarm name | Level |
|---|---|---|---|---|
| 1 | Alarm 1 | 0 | Input string voltage is high | important |
| 2 | Alarm 1 | 1 | DC arc fault * | important |
| 3 | Alarm 1 | 2 | String reverse connection | important |
| 4 | Alarm 1 | 3 | reserve | |
| 5 | Alarm 1 | 4 | reserve | |
| 6 | Alarm 1 | 5 | reserve | |
| 7 | Alarm 1 | 6 | reserve | |
| 8 | Alarm 1 | 7 | Grid power outage | important |
| 9 | Alarm 1 | 8 | Abnormal power grid voltage | important |
| 10 | Alarm 1 | 9 | reserve | |
| 11 | Alarm 1 | 10 | reserve | |
| 12 | Alarm 1 | 11 | Abnormal power grid frequency | important |
| 13 | Alarm 1 | 12 | reserve | |
| 14 | Alarm 1 | 13 | reserve | |
| 15 | Alarm 1 | 14 | Output overcurrent | important |
| 16 | Alarm 1 | 15 | The DC component of the output current is too large | important |

| Number | Alarm | Bit | Alarm name | Level |
|---|---|---|---|---|
| 17 | Alarm 2 | 0 | Abnormal residual current | important |
| 18 | Alarm 2 | 1 | System grounding abnormality | important |
| 19 | Alarm 2 | 2 | Low insulation resistance | important |
| 20 | Alarm 2 | 3 | Temperature is too high | important |
| 21 | Alarm 2 | 4 | reserve | |
| 22 | Alarm 2 | 5 | reserve | |
| 23 | Alarm 2 | 6 | reserve | |
| 24 | Alarm 2 | 7 | reserve | |
| 25 | Alarm 2 | 8 | reserve | |

| | | | | |
|---|---|---|---|---|
| 26 | Alarm 2 | 9 | Energy storage equipment abnormality | important |
| 27 | Alarm 2 | 10 | isolated island | important |
| 28 | Alarm 2 | 11 | reserve | |
| 29 | Alarm 2 | 12 | reserve | |
| 30 | Alarm 2 | 13 | reserve | |
| 31 | Alarm 2 | 14 | Off-grid output overload * | important |
| 32 | Alarm 2 | 15 | reserve | |

| Number | Alarm | Bit | Alarm name | Level |
|---|---|---|---|---|
| 33 | Alarm 3 | 0 | reserve | |
| 34 | Alarm 3 | 1 | reserve | |
| 35 | Alarm 3 | 2 | reserve | |
| 36 | Alarm 3 | 3 | External fan abnormality | important |
| 37 | Alarm 3 | 4 | Energy storage reverse connection * | important |
| 38 | Alarm 3 | 5 | reserve | |
| 39 | Alarm 3 | 6 | reserve | |
| 40 | Alarm 3 | 7 | reserve | |
| 41 | Alarm 3 | 8 | reserve | |
| 42 | Alarm 3 | 9 | Meter Lost | |
| 43 | Alarm 3 | 10 | BMS Lost | |
| 44 | Alarm 3 | 11 | reserve | |
| 45 | Alarm 3 | 12 | reserve | |
| 46 | Alarm 3 | 13 | reserve | |
| 47 | Alarm 3 | 14 | reserve | |
| 48 | Alarm 3 | 15 | reserve | |

**Notice**

The above table is a summary of FOX inverter alarm information. Some alarms need to be equipped with corresponding function modules before they can be detected.

## 4.2 Grid standard code

**Table 4-2** List of power grid standard codes

| Enumeration value | Standard name | Applicable countries or regions |
|---|---|---|
| 0 | AS4777_AU | Australia |
| 1 | AS4777_NZ | New Zealand |
| 2 | G98_UK | U.K. |
| 3 | G99_UK | U.K. |
| 4 | EN50549_NL | Netherlands |
| 5 | CEI021_A | Italy |
| 6 | VDE0126 | Germany |
| 7 | VDE4105_DE | Germany |
| 8 | NBR-220_BR | Brazil |
| 9 | NBR-240_BR | Brazil |
| 10 | IEC61727 | India |
| 11 | Philippines | Philippines |
| 12 | NRS_SA | South Africa |
| 13 | Vietnam | Vietnam |
| 14 | EN50549_PL_A | Poland |
| 15 | EN50549_PT | Portugal |
| 16 | PPDS_CR | Czech Republic |
| 17 | UNE-206_SP | Spain |
| 18 | RD1699_SP | Spain |
| 19 | Belgium | Belgium |
| 20 | VFR2019_FR | France |
| 21 | UTE_FR | France |
| 22 | Singapore | Singapore |
| 23 | Indonesia | Indonesia |
| 24 | Malaysia | Malaysia |
| 25 | Cambodia | Cambodia |
| 26 | PEA_TH | Thailand |
| 27 | MEA_TH | Thailand |

| Enumeration value | Standard name | Applicable countries or regions |
|---|---|---|
| 28 | Sri Lanka | Sri Lanka |
| 29 | Pakistan | Pakistan |
| 30 | Ireland-Micro | Ireland |
| 31 | Denmark 3.2.1 | Denmark |
| 32 | Slovakia | Slovakia |
| 33 | Austria | Austria |
| 34 | Switzerland | Switzerland |
| 35 | Slovenia | slovenia |
| 36 | Hungary | Hungary |
| 3 7 | Serbia | Serbia |
| 3 8 | Croatia | Croatia |
| 39 | Turkey | Türkiye |
| 40 | Cyprus | Cyprus |
| 41 | Bulgaria | Bulgaria |
| 42 | Romania | Romania |
| 43 | Greece | Greece |
| 44 | Latvia | Latvia |
| 45 | Lithuania | Lithuania |
| 46 | Estonia | Estonia |
| 47 | Sweden | Sweden |
| 48 | Norway | Norway |
| 49 | Finland | Finland |
| 50 | Argentina | Argentina |
| 51 | Chile BT | Chile |
| 52 | Mexico | Mexico |
| 53 | USA-120/240 | USA |
| 54 | Hawaii-120/240 | Canada |
| 55 | CQC_CN | China |
| 56 | Japan | Japan |
| 57 | CQC_CN-1 | China (wide range) |
| 58 | Local | India (wide range) |

| Enumeration value | Standard name | Applicable countries or regions |
|---|---|---|
| 59 | Saudi Arabia | Saudi Arabia |
| 60 | AS4777_AU-2020A | Australia(A) |
| 61 | AS4777_AU-2020B | Australia(B) |
| 62 | AS4777_AU-2020C | Australia(C) |
| 63 | AS4777_NZ-2020 | New Zealand |
| 64 | CQC_CN-2 | China (wide range 2) |
| 65 | CEI021_B | Italy |
| 6 6 | CEI021_Areti_A | Italy |
| 6 7 | CEI021_Areti_B | Italy |
| 6 8 | NBR-220/380 | Brazil |
| 6 9 | Spain | Spain |
| 70 | CQC_CN-3 | China |
| 71 | PR-120/240 | Puerto Rico |
| 72 | G98_NI | Northern Ireland |
| 73 | G99_NI | Northern Ireland |
| 74 | USA-120/208 | USA |
| 75 | VDE4110_DE | Germany |
| 76 | KSC8564 | South Korea |
| 77 | KSC8565 | South Korea |
| 78 | PR-LUMA-120/240 | Puerto Rico |
| 79 | CEI016 | Italy |
| 80 | DUBAI | Dubai |
| 81 | Denmark3.2.2 | Denmark |
| 82 | TR 3.3.1-DK1 | Denmark |
| 83 | TR 3.3.1-DK2 | Denmark |
| 84 | Chile MT-A | Chile |
| 85 | Chile MT-B | Chile |
| 86 | EN50549_FR | France |
| 87 | NBR-127/220 | Brazil |
| 88 | NBR-220/380_W | Brazil |

| Enumeration value | Standard name | Applicable countries or regions |
|---|---|---|
| 89 | TWN-T | Taiwan, China |
| 90 | TWN-S | Taiwan, China |
| 91 | Israel | Israel |
| 92 | EN50549_FR_W | France |
| 93 | Test-50Hz | General |
| 94 | Test-60Hz | General |
| 95 | CQC_CN-4 | China |
| 96 | EN 50549-2 | Europe |
| 97 | CQC_CN-5 | China |
| 98 | Austria-B | Austria |
| 99 | Qatar-LV | Qatar |
| 100 | USA-120 | USA |
| 101 | USA-277/480 | USA |
| 102 | Jamaica-110 | Jamaica |
| 103 | TpyeB-DK1 | Denmark |
| 104 | TpyeB-DK2 | Denmark |
| 105 | Ireland-Mini | Ireland |
| 106 | EN50549_PL_B | Poland |
| 107 | PPDS_CR_B | Czech Republic |
| 108 | Colombia | Colombia |
| 109 | Austria_A_North | Austria |
| 110 | Canada-120/240 | Canada |
| 111 | PR-LUMA-120/208 | Puerto Rico |
| 112 | PR-LUMA-277/480 | Puerto Rico |
| 113 | Ukraine | Ukraine |
| | | |
| | | |
| | | |
| | | |
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| | | |
|---|---|---|
| | | |
| | | |
| | | |

**Notice**

The setting of the power grid standard code needs to be selected according to local regulatory requirements.

# 5 Communication protocol overview

The Modbus communication protocol is divided into the following layers (Physical layer, Data link layer, and Application layer), which are described layer by layer:

| Application layer |
| --- |
| Data link layer |
| Physical layer |

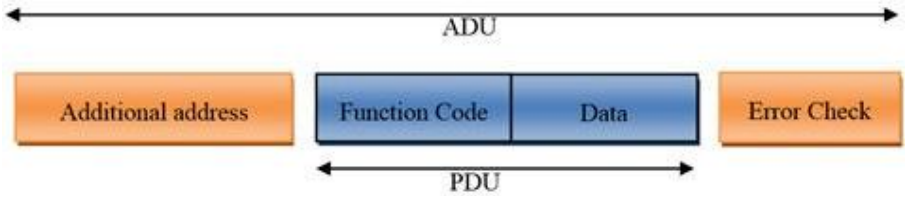
## 5.1 Physical layer

FoxESS inverter provides Modbus communication based on RS485 physical layer medium. Comply with Modbus-RTU format convention.

## 5.2 Data link layer

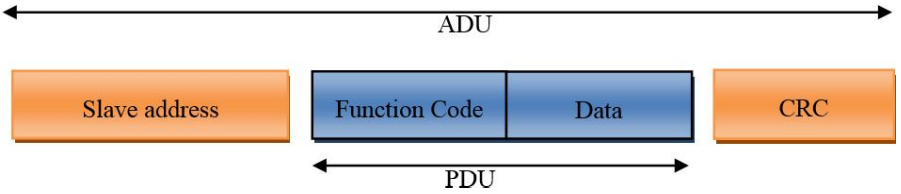
The general frame structure of the Modbus protocol is as follows:

**Figure 5-2** Modbus general frame format




### 5.2.1 Modbus-RTU

**Figure 5-3** Modbus-RTU frame format




#### 5.2.1.1 ADU length

Based on the serial bus ADU is designed according to 256 bytes, where:

Slave address : 1 byte

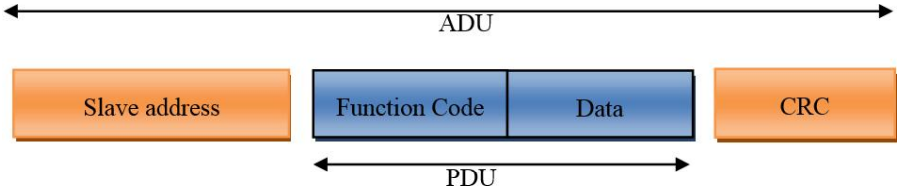
CRC : 2 bytes

PDU: : 253 bytes

### 5.2.1.2 Mailing address

Modbus-RTU communication is commonly used in serial communication . Slave address represents the slave address.

**Figure 5-4 Modbus** general frame format




The address ranges are allocated as follows:

**Table 5-1** Serial link address allocation

| Broadcast address | Slave node address | Reserve |
|---|---|---|
| 0 | 1 ~ 247 | 248 ～ 255 |

The address is reserved for use as a control access to the communication extension module object. FoxESS reserves the right to distribute and use it uniformly.

### 5.2.1.3 CRC check

The CRC check range is the check of all bytes before the CRC field, using 16-bit CRC check. The implemented reference code is as follows:

**High-Order Byte Table**

```
/* Table of CRC values for high–order byte */
static unsigned char auchCRCHi[] = {
    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81,
    0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
    0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01,
    0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81,
    0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
    0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01,
    0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81,
    0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
    0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01,
    0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
    0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81,
    0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
    0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01,
    0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81,
    0x40
};
```

**Low-Order Byte Table**

```
/* Table of CRC values for low–order byte */
static char auchCRCLo[] = {
   0x00, 0xC0, 0xC1, 0x01, 0xC3, 0x03, 0x02, 0xC2, 0xC6, 0x06, 0x07, 0xC7, 0x05, 0xC5, 0xC4,
   0x04, 0xCC, 0x0C, 0x0D, 0xCD, 0x0F, 0xCF, 0xCE, 0x0E, 0x0A, 0xCA, 0xCB, 0x0B, 0xC9, 0x09,
   0x08, 0xC8, 0xD8, 0x18, 0x19, 0xD9, 0x1B, 0xDB, 0xDA, 0x1A, 0x1E, 0xDE, 0xDF, 0x1F, 0xDD,
   0x1D, 0x1C, 0xDC, 0x14, 0xD4, 0xD5, 0x15, 0xD7, 0x17, 0x16, 0xD6, 0xD2, 0x12, 0x13, 0xD3,
   0x11, 0xD1, 0xD0, 0x10, 0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3, 0xF2, 0x32, 0x36, 0xF6, 0xF7,
   0x37, 0xF5, 0x35, 0x34, 0xF4, 0x3C, 0xFC, 0xFD, 0x3D, 0xFF, 0x3F, 0x3E, 0xFE, 0xFA, 0x3A,
   0x3B, 0xFB, 0x39, 0xF9, 0xF8, 0x38, 0x28, 0xE8, 0xE9, 0x29, 0xEB, 0x2B, 0x2A, 0xEA, 0xEE,
   0x2E, 0x2F, 0xEF, 0x2D, 0xED, 0xEC, 0x2C, 0xE4, 0x24, 0x25, 0xE5, 0x27, 0xE7, 0xE6, 0x26,
   0x22, 0xE2, 0xE3, 0x23, 0xE1, 0x21, 0x20, 0xE0, 0xA0, 0x60, 0x61, 0xA1, 0x63, 0xA3, 0xA2,
   0x62, 0x66, 0xA6, 0xA7, 0x67, 0xA5, 0x65, 0x64, 0xA4, 0x6C, 0xAC, 0xAD, 0x6D, 0xAF, 0x6F,
   0x6E, 0xAE, 0xAA, 0x6A, 0x6B, 0xAB, 0x69, 0xA9, 0xA8, 0x68, 0x78, 0xB8, 0xB9, 0x79, 0xBB,
   0x7B, 0x7A, 0xBA, 0xBE, 0x7E, 0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4, 0x74, 0x75, 0xB5,
   0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71, 0x70, 0xB0, 0x50, 0x90, 0x91,
   0x51, 0x93, 0x53, 0x52, 0x92, 0x96, 0x56, 0x57, 0x97, 0x55, 0x95, 0x94, 0x54, 0x9C, 0x5C,
   0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E, 0x5A, 0x9A, 0x9B, 0x5B, 0x99, 0x59, 0x58, 0x98, 0x88,
   0x48, 0x49, 0x89, 0x4B, 0x8B, 0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D, 0x4D, 0x4C, 0x8C,
   0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42, 0x43, 0x83, 0x41, 0x81, 0x80,
   0x40
};
```

**CRC Generation Function**

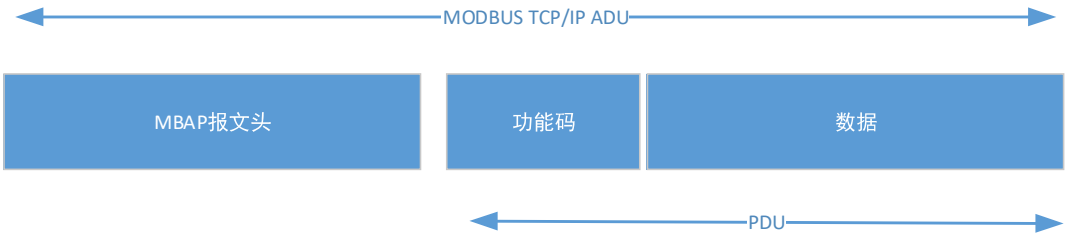
```
unsigned short CRC16 ( puchMsg, usDataLen )          /* The function returns the CRC as a unsigned short type  */
unsigned char *puchMsg ;                             /* message to calculate CRC upon                          */
unsigned short usDataLen ;                           /* quantity of bytes in message                           */
{
        unsigned char uchCRCHi = 0xFF ;              /* high byte of CRC initialized                           */
        unsigned char uchCRCLo = 0xFF ;              /* low byte of CRC initialized                            */
        unsigned uIndex ;                            /* will index into CRC lookup table                       */

        while (usDataLen--)                          /* pass through message buffer                            */
        {
                uIndex = uchCRCLo ^ *puchMsg++ ;     /* calculate the CRC                                      */
                uchCRCLo = uchCRCHi ^ auchCRCHi[uIndex] ;
                uchCRCHi = auchCRCLo[uIndex] ;
        }
        return (uchCRCHi << 8 | uchCRCLo) ;
}
```

Code source: "MODBUS over Serial Line Specification and Implementation Guide V1.02"

### 5.2.2 Modbus- TCP

**Figure 5-5** Modbus- TCP frame




#### 5.2.2.1 ADU length

The standards-based recommended frame length is 260 bytes. When applying some extended functions, the data can extend the ADU to an appropriate length according to its own resource conditions to improve network transmission efficiency. The actual ADU length is reflected in the MBAP header length field.

#### 5.2.2.2 MBAP message header

When Modbus is applied on TCP/IP, a special MBAP header (Modbus application protocol header) will be used to identify the Modbus application data unit (ADU). The MBAP header is divided into 4 fields and a total of 7 words. section, defined as follows:

**Table 5-2 MBAP definition table**

| Data field | Length (Byte) | Description | Client | Server |
|---|---|---|---|---|
| Transport identifier | 2 | Request frame and response frame matching identifier | Client allocation, it is recommended that each frame of data request transmission identifier be different | The identifier in the server-side response frame must be consistent with the request frame |
| Agreement type | 2 | 0 = MODBUS protocol | Client allocation, default is 0 | The identifier in the server-side response frame must be consistent with the request frame |
| Data length | 2 | Subsequent data length | The client allocates based on actual frames | Server allocates based on actual frame length |
| Logical device ID | 1 | 0 | Client requests allocation based on actual frames | The identifier in the server-side response frame must be consistent with the request frame |

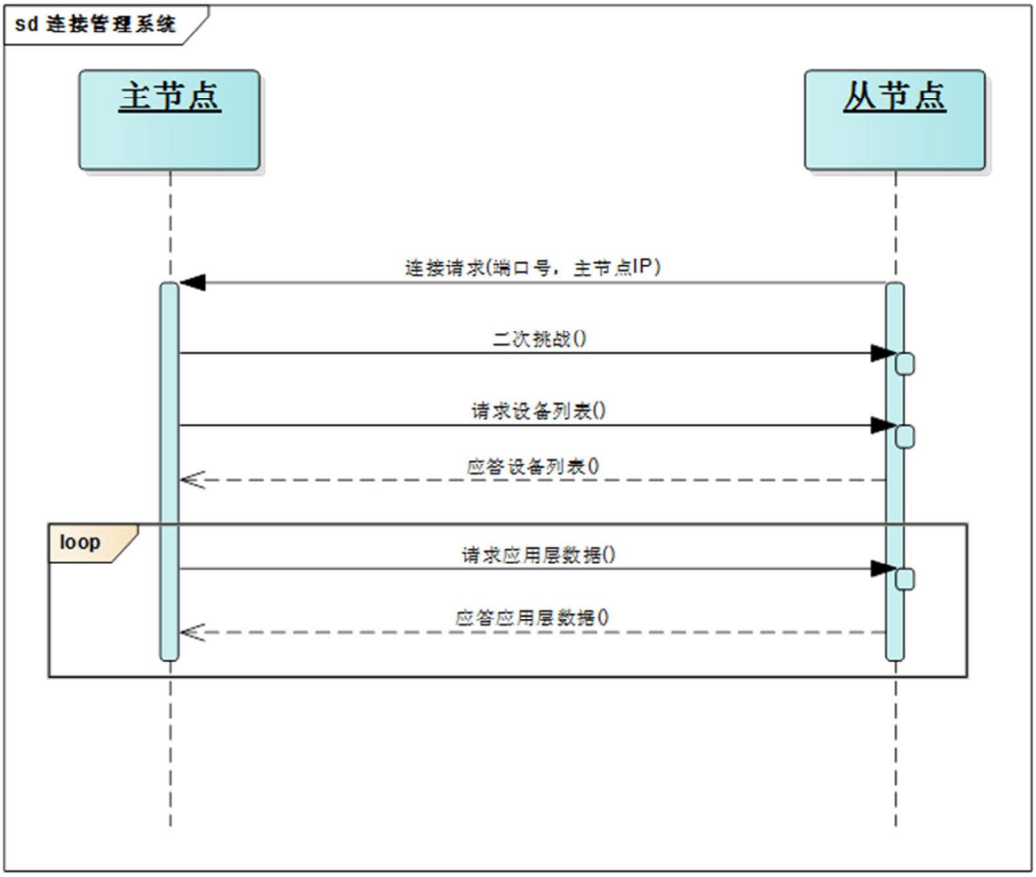
#### 5.2.2.3 TCP port

In a LAN or VPN environment, the master node can actively initiate TCP socket link establishment to the slave node. The master node can use port 502 to request data services from the slave node.
In a cross-public network non-VPN environment, the device deployed on the internal network side needs to initiate a TCP socket link establishment to the master node exposed on the public network side. At this time, you need to preset the fixed access port number of the master node on the slave node side. Based on security and traffic streamlining considerations, the master node is required to provide at least 1 encrypted port and 1 non-encrypted port.

#### 5.2.2.4 TCP link building process

Considering the particularity of cross-public network applications, this section only focuses on describing this scenario. The following figure describes the slave node access process:

**Figure 5-7 TCP secure link establishment process**




## 5.3 Application layer

### 5.3.1 Function code list

**Table 5-2** Function code list

| Function code | Meaning | Remark |
|---|---|---|
| 0x03 | Read register | Supports single and multiple register sequential reads |
| 0x06 | Write to a single register | Support single register write action |
| 0x10 | Write multiple registers | Supports continuous writing of multiple registers |

### 5.3.2 Exception code list

Each network element type needs to ensure that the exception code of its product is unique, and the

name and description are provided uniformly (Chinese and English description information needs to be provided in the network element interface document). Multiple versions of the same network element type must be forward compatible, and the encoding that has been used cannot be reused for others.

**Table 5-3** Summary table of exception codes returned by network elements (0x00-0x8F is the public exception code segment)

| Code | Name | Meaning |
|---|---|---|
| 0x01 | illegal function | For the server (or slave), the function code received in the query is an impermissible operation. This may be because the function code only applies to new equipment and is not available in the selected unit. It is also indicated that the server (or slave) handles this request in an error state, for example because it is unconfigured and requires a return register value. |
| 0x02 | Illegal data address | The query request received by the server contains a disallowed register address. More specifically, the combination of register starting address and register number is invalid. For a controller with 100 registers, the first PDU address is 0, and the last one is 99. If the starting register address in a request is 96 and the number of registers is 4, then this request can get the return value of registers 96, 97, 98, 99. If a request is submitted with a starting register address of 96 and the number of registers is 5, then the request will fail and the exception code 0x02 "illegal data address" returned, because it is trying to read registers 96, 97, 98, 99, 100, But 100 is an address that has no actual definition . |
| 0x03 | illegal data value | The value included in the query is not allowed for the server (or slave). This value indicates a failure in the remaining structure of the combined request, for example: the implicit length is incorrect. It does not mean that, because the MODBUS protocol is not aware of the significance of any special value in any special register, the data item submitted for storage in the register has a value other than that expected by the application. |
| 0x04 | Slave node device failure | During execution, the server failed. |
| 0x06 | slave device busy | The server cannot accept the MODBUS request PDU. It is the client application's responsibility to decide if and when to resend the request. |
| 0x80 | No permission | Authentication fails or permissions expire after timeout, and operation is prohibited. |

### 5.3.3 Read register (0x03)

#### 5.3.3.1 Master node request frame format

**Table 5-4** Master node request frame format

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x03 |
| Register starting address | 2 bytes | 0x0000~0xFFFF |
| Number of registers | 2 bytes | 1~125 |

#### 5.3.3.2 Normal response frame format from the node

**Table 5-5** Normal response frame format of the slave node

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x03 |
| Number of bytes | 1 byte | 2×N |
| Register value | 2×N bytes | NA |

**illustrate**

N is the number of registers.

#### 5.3.3.3 Slave node exception response frame format

**Table 5-6** Slave node exception response frame format

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x83 |
| Exception code | 1 byte | See **exception code list** |

### 5.3.4 Write single register (0x06)

#### 5.3.4.1 Master node request frame format

**Table 5-7** Master node request frame format

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x06 |
| Register address | 2 bytes | 0x0000~0xFFFF |
| Register value | 2 bytes | 0x0000~0xFFFF |

#### 5.3.4.2 Normal response frame format from the node

**Table 5-8** Normal response frame format of the slave node

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x06 |
| Register address | 2 bytes | 0x0000~0xFFFF |
| Register value | 2 bytes | 0x0000~0xFFFF |

#### 5.3.4.3 Slave node exception response frame format

**Table 5-9** Slave node exception response frame format

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x86 |
| Exception code | 1 byte | See **exception code list** |

### 5.3.5 Write multiple registers (0x10)

#### 5.3.5.1 Master node request frame format

**Table 5-10** Master node request frame format

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x10 |
| Register starting address | 2 bytes | 0x0000~0xFFFF |
| Number of registers | 2 bytes | 0x0000~0x007B |
| Number of bytes | 1 byte | 2×N |
| Register value | 2×N bytes | Value |

**illustrate**

N is the number of registers.

#### 5.3.5.2 Normal response frame format from the node

**Table 5-11** Normal response frame format of the slave node

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x10 |
| Register address | 2 bytes | 0x0000~0xFFFF |
| Number of registers | 2 bytes | 0x0000~0x007B |

#### 5.3.5.3 Slave node exception response frame format

**Table 5-12** Slave node exception response frame format

| Data field | Length | Description |
|---|---|---|
| Function code | 1 byte | 0x90 |
| Exception code | 1 byte | See **exception code list** |

### 5.3.6 Configure slave address

This function is used to configure the MODBUS address .

The data frame is composed as follows :

| Data field | Value | Description |
|---|---|---|
| Slave address | 0xF9 | Configure the slave address exclusively |
| Function code | 0x06 | Write to a single register |
| Register address | 0x5A5A | special address |
| Register value | 2 bytes | Write slave address<br>Range: 10 - 99 |
| CRC 16 | 2 bytes | |

## Change record table

| version number | change log | date |
|---|---|---|
| V1.0 | First edition | 2023.07.18 |
| V1.01.00.00 | Add register definition and Modbus-TCP definition | 2023.11.08 |
| V1.02.00.00 | Add some registers to optimise document formatting | 2023.12.06 |
| V1.03.00.00 | Add some registers | 2023.12.29 |
| V1.04.00.00 | Add some registers, and the meter lost information. | 2024.03.20 |
| V1.05.00.00 | 1. Add register 30032, 36100, 36116, 36132, 36148, 36200, 36216, 36232, 36248, 37635, 38333, and 45002, etc.<br>2. Modify the definition information for registers 37632, 38330, 39118, etc.<br>3. Add Alarm 43 information. | 2024.05.16 |
| V1.05.01.00 | 1. Modify the defined Bits 3:2 for reguster 46001. | 2024.05.17 |
| V1.05.02.00 | 1. Add register 37636, 38334, 45003, 45005, 45006, 45007, 49232, 49240, 49241, 49242, 49243, 49244, 49245, etc.<br>2. Modfiy the definition information for registers 37619, 37620, 38317, 38318, 46001, etc. | 2024.08.15 |
| V1.05.03.00 | 1. Add register 39423, 49008, 49010, 49247, 49248, 49249, etc.<br>2. Add register table 3-11. | 2025.01.15 |
| V1.05.04.00 | 1. Modify the definition information for registers 46607, 46608, 46501 to 46514, 46618, 46619, etc.<br>2. Add register 46620, 49251, 39633, 39635, 39637, 39639, etc.<br>3. Extern the time period table from 24 time periods to 96 time periods, and add comments.<br>4. Add register table 3-13. | 2025.12.15 |
| V1.05.05.00 | 1. Mark these registers 39633 to 39640 as not supported.<br>2. Modify the Bit-9 defined of register 46001.<br>3. Modify the signal name of register 49210.<br>4. Add the register 39424, 39425, 39426, 46026, 49252, 49253, 49254, 49255, 49256, 49257, 49259, 49501, 49503, 49504, 49505, etc.<br>5. Update grid standard code table 4-2. | 2026.07.01 |
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